Abstract

With the exponential growth in digital banking, online transactions have become an integral part of modern financial systems. However, this rapid digitalization also brings the growing threat of fraudulent activities, causing severe financial losses and damaging customer trust. Traditional fraud detection systems are often slow, reactive, and incapable of handling large volumes of real-time data. This project proposes a real-time fraud detection system using Apache Kafka for real-time data streaming and Machine Learning (ML) algorithms for intelligent pattern recognition. By integrating Kafka's distributed messaging framework with ML models trained on historical transaction data, the system aims to detect anomalies in transaction behavior as they occur. This setup provides a scalable, fault-tolerant, and efficient mechanism to detect and respond to fraud in real-time, enhancing security and user trust in digital banking services.